Brent Minchew

Professor of Geophysics
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Curriculum Vitae

Bio

I study how glaciers and ice sheets respond to climate change, with emphasis on the fundamental mechanisms that govern rates of sea-level rise in my daughter's lifetime. My research leverages satellite remote sensing, mathematical theory, and computational models to study the flow, deformation, and fracture of glacier ice across scales from atomic to continental and seconds to centuries. My work allows me to see nature at its most elegant, both fragile and resilient, stately and responsive.

Education

2016	Ph.D. Geophysics, California Institute of Technology
2010	M.S. Aerospace Engineering, University of Texas at Austin
2008	B.S. Aerospace Engineering, University of Texas at Austin

Academic Positions

2025–	Professor of Geophysics, California Institute of Technology
2024–2025	Chair of the Program in Geophysics, Massachusetts Institute of Technology
2024–2025	Associate Professor (with tenure), Massachusetts Institute of Technology
2023–2024	Associate Professor (without tenure), Massachusetts Institute of Technology
2018–2023	Assistant Professor, Massachusetts Institute of Technology
2016–2018	NSF Postdoctoral Fellow, British Antarctic Survey
2010–2015	Graduate Research and Teaching Assistant, California Institute of Technology
2009 & 2010	Graduate Research Fellow, NASA Jet Propulsion Laboratory
2006–2010	Research and Teaching Assistant, University of Texas at Austin

Nonacademic Positions

2024–	Co-Founder and Chief Scientist, Arête Glacier Initiative
1996–2004	Staff Sergeant, United States Marine Corps

Select Awards and Honors

2022	Class of 1948 Career Development Professorship, MIT
2019	Cecil and Ida Green Career Development Chair, MIT

2014	NSF Earth Sciences Postdoctoral Fellowship
2013–2015	ARCS and Albert Parvin Foundation Fellowship
2012	IEEE Transactions on Geoscience and Remote Sensing Editor's Choice Award
2011–2014	NASA Earth and Space Sciences Fellowship
2011–2013	ARCS Foundation Fellowship

Publications and Presentations

Scholarly articles

(46 in total; * indicates students and * postdocs and other group members)

46. S. Wells-Moran, **B. M. Minchew**, and B. V. Riel. "Near-total loss of buttressing stresses observed on Pine Island Ice Shelf, West Antarctica". *Proceedings of the National Academy of Sciences* 123.33 (2026), e2602994123. DOI: 0.1073/pnas.2602994123.
45. C. R. Meyer, K. L. P. Warburton, A. N. Sommers, and **B. M. Minchew**. "A model of water extraction from the subglacial hydrologic system under idealized conditions". *The Cryosphere* 20.5 (2026), pp. 2659–2680. DOI: 10.5194/tc-20-2659-2026.
44. D. F. Martin, S. B. Kachuck, J. D. Millstein, and **B. M. Minchew**. "Rates of sea-level rise are highly sensitive to ice viscosity parameters in model benchmarks". *AGU Advances* 7.2 (2026), e2025AV001946. DOI: 10.1029/2025AV001946.
43. F. M. Elgart*, **B. M. Minchew**, and C. R. Meyer. "Estimating effective elasticity in grounding zones of the Ross Ice Shelf with tidal flexure from ICESat-2". *Journal of Glaciology* 72 (2026), e23. DOI: 10.1017/jog.2026.10137.
42. C. R. Meyer, A. G. Stubblefield, **B. M. Minchew**, S. S. Pegler, A. Berne, J. J. Buffo, and T. C. Tomlinson. "Implications of shallow-shell models for topographic relaxation on icy satellites". *Journal of Geophysical Research - Planets* 130.12 (2025), e2025JE009247. DOI: 10.1029/2025JE009247.
41. N. Narayanan*, A. Sommer, W. Chu, J. Steiner, M. A. Siddique, C. R. Meyer, and **B. M. Minchew**. "Simulating seasonal evolution of subglacial hydrology at a surging glacier in the Karakoram". *Journal of Glaciology* 71 (2025), e94. DOI: 10.1017/jog.2025.10078.
40. S. Wells-Moran**, M. Ranganathan*, and **B. M. Minchew**. "Fracture criteria and tensile strength for natural glacier ice calibrated from remote sensing observations of Antarctic ice shelves". *Journal of Glaciology* 71 (2025), e47. DOI: 10.1017/jog.2024.104.
39. U. Ovienmhada*, M. Hines-Shanks, M. Krisch, A. Diongue, **B. M. Minchew**, and D. Wood. "Spatiotemporal Facility-Level Patterns of Summer Heat Exposure, Vulnerability, and Risk in United States Prison landscapes". *GeoHealth* 8.9 (2024). DOI: 10.1029/2024GH001108.
38. M. Ranganathan* and **B. M. Minchew**. "A modified viscous flow law for natural glacier ice: Scaling from laboratories to ice sheets". *Proceedings of National Academy of Sciences* 121.23 (2024). DOI: 10.1073/pnas.2309788121.

37. F. Clerc*, M. D. Behn, and **B. M. Minchew**. “Deglaciation-enhanced mantle CO2 fluxes at Yellowstone imply positive climate feedbacks”. *Nature Communications* 15.1526 (2024), pp. 1–10. DOI: `10.1038/s41467-024-45890-z`.
36. Y. Sun*, B. V. Riel⋆, and **B. M. Minchew**. “Disintegration and Buttressing Effect of the Landfast Sea Ice in the Larsen B Embayment, Antarctic Peninsula”. *Geophysical Research Letters* 50.16 (2023). DOI: `10.1029/2023GL104066`.
35. M. Ranganathan*, J. W. Barotta**, C. R. Meyer, and **B. M. Minchew**. “Meltwater generation in ice stream shear margins: Case study in Antarctic ice streams”. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 479.2273 (2023). DOI: `10.1098/rspa.2022.0473`.
34. B. V. Riel⋆ and **B. M. Minchew**. “Variational inference of ice shelf rheology with physics-informed machine learning”. *Journal of Glaciology* 69.277 (2023), pp. 1167–1186. DOI: `10.1017/jog.2023.8`.
33. M. Zhong, M. Simons, **B. M. Minchew**, and L. Zhu. “Inferring tide-induced ephemeral grounding and subsequent dynamical response in an ice-shelf-stream system: Rutford Ice Stream, West Antarctica”. *Journal of Geophysical Research – Earth Surface* 128.2 (2023). DOI: `10.1029/2022JF006789`.
32. L. Ultee⋆, D. Felikson, **B. M. Minchew**, L. A. Stearns, and B. V. Riel⋆. “Helheim Glacier ice velocity variability responds to runoff and terminus position change at different timescales”. *Nature Communications* 13.6022 (2022). DOI: `10.1038/s41467-022-33292-y`.
31. J. D. Millstein*, **B. M. Minchew**, and S. S. Pegler. “Ice viscosity is more sensitive to stress than commonly assumed”. *Nature Communications Earth and Environment* 3.57 (2022). DOI: `10.1038/s43247-022-00385-x`.
30. M. Ranganathan*, **B. M. Minchew**, C. R. Meyer, and M. Pec. “Recrystallization of ice enhances creep and the vulnerability to fracture of ice shelves”. *Earth and Planetary Science Letters* 576 (2021). DOI: `10.1016/j.epsl.2021.117219`.
29. B. V. Riel⋆, **B. M. Minchew**, and T. Bischoff. “Data-driven inference of the mechanics of slip along glacier beds using physics-informed neural networks: Case study on Rutford Ice Stream, Antarctica”. *Journal of Advances in Modeling Earth Systems* 13.11 (2021), e2021MS002621. DOI: `10.1029/2021MS002621`.
28. G. Guerin**, A. Mordret⋆, D. Rivet, B. P. Lipovsky, and **B. M. Minchew**. “Frictional origin of slip events of the Whillans Ice Stream, Antarctica”. *Geophysical Research Letters* 48.11 (2021), pp. 1–9. DOI: `10.1029/2021GL092950`.
27. K. S. Shah*, S. S. Pegler, and **B. M. Minchew**. “Dynamics of two-layer fluid flows on inclined surfaces”. *Journal of Fluid Mechanics* 917.A54 (2021), pp. 1–33. DOI: `10.1017/jfm.2021.273`.
26. P. Hunter*, C. R. Meyer, **B. M. Minchew**, M. Haseloff, and A. Rempel. “Thermal controls on ice stream shear margins”. *Journal of Glaciology* 67.263 (2021), pp. 435–449. DOI: `10.1017/jog.2020.118`.
25. B. V. Riel⋆, **B. M. Minchew**, and I. Joughin. “Observing traveling waves in glaciers with remote sensing: New flexible time-series methods and application to Sermeq Kujalleq (Jakobshavn Isbræ), Greenland”. *The Cryosphere* 15.1 (2021), pp. 407–429. DOI: `10.5194/tc-15-407-2021`.

24. M. Ranganathan*, **B. M. Minchew**, C. R. Meyer, and G. H. Gudmundsson. “A new approach to inferring basal drag and ice rheology in ice streams, with applications to West Antarctic ice streams”. *Journal of Glaciology* 67.262 (2021). DOI: `10.1017/jog.2020.95`.
23. E. H. Ultee*, C. R. Meyer, and **B. M. Minchew**. “Tensile strength of glacial ice deduced from observations of the 2015 Eastern Skaftá Cauldron collapse, Vatnajökull ice cap, Iceland”. *Journal of Glaciology* 66.260 (2020), pp. 1024–1033. DOI: `10.1017/jog.2020.65`.
22. **B. M. Minchew** and C. R. Meyer. “Dilation of subglacial sediment governs incipient surge motion in glaciers with deformable beds”. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 476.2238 (2020). DOI: `10.1098/rspa.2020.0033`.
21. **B. M. Minchew** and I. Joughin. “Toward a universal glacier slip law”. *Science* 368.6486 (2020), pp. 29–30. DOI: `10.1126/science.abb3566`.
20. F. Clerc*, **B. M. Minchew**, and M. D. Behn. “Marine ice cliff instability mitigated by slow removal of ice shelves”. *Geophysical Research Letters* 46.21 (2019), pp. 12108–12116. DOI: `10.1029/2019GL084183`.
19. **B. M. Minchew**, C. R. Meyer, S. S. Pegler, B. P. Lipovsky, A. W. Rempel, G. H. Gudmundsson, and N. R. Iverson. “Comment on: “Friction at the bed does not control fast glacier flow” by L. A. Stearns and C. J. van der Veen”. *Science* 363.6427 (2019). DOI: `10.1126/science.aau6055`.
18. C. R. Meyer, A. Yehya, **B. M. Minchew**, and J. R. Rice. “A model for the downstream evolution of temperate ice and subglacial hydrology along ice stream shear margins”. *Journal of Geophysical Research - Earth Surface* 123.8 (2018), pp. 1682–1698. DOI: `10.1029/2018JF004669`.
17. C. R. Meyer and **B. M. Minchew**. “Temperate ice in the shear margins of the Antarctic Ice Sheet: controlling processes and preliminary locations”. *Earth and Planetary Science Letters* 498 (2018), pp. 17–26. DOI: `10.1016/j.epsl.2018.06.028`.
16. **B. M. Minchew**, C. R. Meyer, A. A. Robel, G. H. Gudumundsson, and M. Simons. “Processes controlling the downstream evolution of ice rheology in glacier shear margins: Case study on Rutford Ice Stream, West Antarctica”. *Journal of Glaciology* 64.246 (2018), pp. 583–594. DOI: `10.1017/jog.2018.47`.
15. **B. M. Minchew**, G. H. Gudmundsson, A. Gardner, F. S. Paolo, and H. A. Fricker. “Modeling the dynamic response of outlet glaciers to observed ice-shelf thinning in the Bellingshausen Sea Sector, West Antarctica”. *Journal of Glaciology* 64.244 (2018), pp. 333–342. DOI: `10.1017/jog.2018.24`.
14. S. Angelliaume, P. Dubois-Fernandez, C. E. Jones, B. Holt, **B. M. Minchew**, E. Amri, and V. Miegebielle. “SAR imagery for detecting sea surface slicks: Performance assessment of polarimetric parameters”. *IEEE Transactions on Geoscience and Remote Sensing* 56.8 (2018), pp. 4237–4257. DOI: `10.1109/TGRS.2018.2803216`.
13. A. A. Robel, V. C. Tsai, **B. M. Minchew**, and M. Simons. “Tidal modulation of ice shelf buttressing stresses”. *Annals of Glaciology* 58.74 (2017), pp. 12–20. DOI: `10.1017/aog.2017.22`.

12. P. Milillo, **B. M. Minchew**, M. Simons, P. Agram, and B. Riel. "Geodetic imaging of time-dependent three-component surface deformation: application to tidal-timescale ice flow of Rutford Ice Stream, West Antarctica". *IEEE Transactions on Geoscience and Remote Sensing* 55.10 (2017), pp. 5515–5524. DOI: `10.1109/TGRS.2017.2709783`.
11. S. Angelliaume, **B. M. Minchew**, S. Chatiang, P. Martineau, and V. Miegebielle. "Multifrequency radar imagery and characterization of hazardous and noxious substances at sea". *IEEE Transactions on Geoscience and Remote Sensing* 55.5 (2017). DOI: `10.1109/TGRS.2017.2661325`.
10. **B. M. Minchew**, M. Simons, B. V. Riel, and P. Milillo. "Tidally induced variations in vertical and horizontal motion on Rutford Ice Stream, West Antarctica, inferred from remotely sensed observations". *Journal of Geophysical Research: Earth Surface* 122 (2017), pp. 167–190. DOI: `10.1002/2016JF003971`.
9. **B. M. Minchew**, M. Simons, H. Björnsson, F. Pálsson, M. Morlighem, H. Seroussi, E. Larour, and S. Hensley. "Plastic bed beneath Hofsjökull Ice Cap, central Iceland, and the sensitivity of ice flow to surface meltwater flux". *Journal of Glaciology* 62.231 (2016), pp. 147–158. DOI: `10.1017/jog.2016.26`.
8. P. Milillo, B. Riel, **B. M. Minchew**, S. H. Yun, M. Simons, and P. Lundgren. "On the synergistic use of SAR constellations' data exploitation for earth science and natural hazard response". *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9.3 (2015), pp. 1095–1100. DOI: `10.1109/JSTARS.2015.2465166`.
7. M. J. Collins, M. Denbina, **B. M. Minchew**, C.E. Jones, and B. Holt. "On the use of simulated airborne compact polarimetric SAR for characterizing oil-water mixing of the Deepwater Horizon oil spill". *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 8.3 (2015), pp. 1062–1077. DOI: `10.1109/JSTARS.2015.2401041`.
6. **B. M. Minchew**, M. Simons, S. Hensley, H. Björnsson, and F. Pálsson. "Early melt-season velocity fields of Langjökull and Hofsjökull ice caps, central Iceland". *Journal of Glaciology* 61.226 (2015), pp. 253–266. DOI: `10.3189/2015JoG14J023`.
5. J. S. Scheingross, **B. M. Minchew**, B.H. Mackey, M. Simons, M.P. Lamb, and S. Hensley. "Fault zone controls on the spatial distribution of slow-moving landslides". *GSA Bulletin* 125.3-4 (2013), pp. 473–489. DOI: `10.1130/B30719.1`.
4. **B. M. Minchew**, C.E. Jones, and B. Holt. "Polarimetric analysis of backscatter from the Deepwater Horizon oil spill using L-band synthetic aperture radar". *IEEE Transactions on Geoscience and Remote Sensing* 50.10 (2012), pp. 3812 –3830. DOI: `10.1109/TGRS.2012.2185804`.
3. **B. M. Minchew**. "Determining the mixing of oil and seawater using polarimetric synthetic aperture radar". *Geophysical Research Letters* 39.16 (2012). L16607. DOI: `10.1029/2012GL052304`.
2. V. C. Tsai, **B. M. Minchew**, M. P. Lamb, and J. P. Ampuero. "A physical model for seismic noise generation from sediment transport in rivers". *Geophysical Research Letters* 39.2 (2012). L02404. DOI: `10.1029/2011GL050255`.
1. C. E. Jones, **B. M. Minchew**, B. Holt, and S. Hensley. "Studies of the Deepwater Horizon Oil Spill with the UAVSAR radar". *Monitoring and Modeling the Deepwater Horizon Oil Spill: A Record-Breaking Enterprise*. Vol. 195. Washington, DC: AGU, 2011, pp. 33–50. DOI: `10.1029/2011GM001113`.

Other publications

2. Brent Minchew and Colin Meyer. "We study glaciers. 'Artificial glaciers' and other tech may halt their total collapse". *The Guardian* (2026). Op-ed, originally entitled: "Sea levels are rising rapidly, threatening the homes of millions. Let's do something about it".
1. B. M. Minchew, L. Mahle, P. Ghosh, M. Sistla, and E. N. Martin. "Polar infrastructure and science for national security: A federal agenda to promote glacier resilience and strengthen American competitiveness". *Day One Project, Federation of American Scientists* (2024).

Working manuscripts and preprints

(Group members and advisees: * indicates graduate students, ** undergraduates, and * postdocs and research scientists)

5. A. S. Miller, J. T. Cole, C. R. Meyer, and **B. M. Minchew**. "Models of Thermosyphon Performance, Design, and Influence on Glacier Flow". *submitted* (2026). DOI: `10.31223/X56Z19`.
4. B. Riel and **B. M. Minchew**. "Identifying the mechanisms behind seasonal velocity changes in Greenland glaciers". *submitted* (2026).
3. E. Kramer* and **B. M. Minchew**. "Magnitude-order improvements to the measurement precision of Antarctic grounding zone dynamics using SAR satellite constellations". *in review* (2026).
2. F. M. Elgart* and **B. M. Minchew**. "Estimating Ice Shelf Thickness in Grounding Zones of the Filchner-Ronne Ice Shelf with Tidal Flexure from ICESat-2". *in review* (2026).
1. C. Mouchon*, **B. M. Minchew**, and C. R. Meyer. "Evolution of ice viscosity with stress as constrained by observations of a surge on Vavilov Ice Cap". *in review* (2026).

Invited presentations

(past 5 years only)

B. M. Minchew. "Rheology: The workhorse of glaciology". *Challenges, Advances, and Emerging Directions in Ice Deformation*. Apr. 2026.

B. M. Minchew. "Glacier dynamics from viscous flow to iceberg calving". *UCLA EPSS Colloquium*. Jan. 2026.

B. M. Minchew. "The rapid demise of a major outlet glacier in West Antarctica". *Caltech Geo Club*. Dec. 2025.

B. M. Minchew. "How fast will sea-levels rise and what are we going to do about it?" *Busan IAMAS-IACS-IAPSO Joint Assembly 2025*. 2025.

B. M. Minchew. "Some ideas for slowing glaciers and sea-level rise". *UC Santa Barbara Kavli Institute for Theoretical Physics*. 2025.

B. M. Minchew. "How fast will sea-levels rise and what are we going to do about it?" *MIT EVDT Community Meeting*. May 2025.

B. M. Minchew. "Glacier dynamics from viscous flow to iceberg calving". *Geodynamics Seminar, Woods Hole Oceanographic Institution*. 2025.

B. M. Minchew. “Glacier dynamics from viscous flow to iceberg calving”. *Geodynamics Seminar, Lamont Doherty Earth Observatory, Columbia University*. 2025.

B. M. Minchew and C. R. Meyer. “Between a rock and hard place: Some thoughts on how to approach responsible glacier interventions”. *AGU Fall Meeting*. Dec. 2024.

B. M. Minchew. “Slowing glaciers by increasing drag at the bed: A brief primer”. *Glacial Geoengineering Workshop, Stanford University*. Dec. 2023.

B. M. Minchew. “Glacier dynamics from viscous flow to iceberg calving”. *Geological and Planetary Sciences Division Seminar, California Institute of Technology*. Nov. 2023.

B. M. Minchew. “Principles behind modulating basal slip in fast-flowing glaciers”. *Glacial Geoengineering Workshop, University of Chicago*. Oct. 2023.

B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Geology and Geophysics Department Seminar, Woods Hole Oceanogrpahic Institute*. Aug. 2023.

B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Geophysics Seminar, Stanford University*. Apr. 2023.

B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Department of Earth, Environmental, and Planetary Sciences Seminar, Brown University*. Feb. 2023.

B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Earth and Environmental Science Seminar, University of Pennsylvania*. Feb. 2023.

B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *University of California, Santa Barbara*. Feb. 2023.

B. M. Minchew. “Observing at currently unobservable space and time scales”. *Caltech Seismo Lab Centennial*. Nov. 2022.

B. M. Minchew. “Our evolving understanding of the deformation of glacier ice”. *Tufts University, Civil and Environmental Engineering Colloquium*. Oct. 2022.

B. M. Minchew. “Sea-level rise in the 21st century: Education and innovation driving science and adaptation”. *MIT-Portugal Annual Conference*. Sept. 2022.

B. M. Minchew. “Earth Science in the age of NISAR”. *NISAR Science Community Workshop*. Aug. 2022.

B. M. Minchew. “Managing the risks of sea-level rise through improved ice-sheet monitoring”. *Eureka meets the Atlantic | Boston*. Apr. 2022.

B. M. Minchew. “Flow and deformation of glacier ice”. *Seismolab Seminar, California Institute of Technology*. Mar. 2022.

B. M. Minchew. “Lessons learned from observing Earth’s cryosphere from satellites”. *Lowell Observatory, I Heart Pluto Festival*. Feb. 2022.

Grants

2024–	PI – Grantham Foundation: Reducing the risk of sea-level rise by deliberately slowing glaciers in Antarctica: First steps.
2021–	PI – John W. Jarve (1978) Seed Fund: Improving projections of sea-level rise by (deep) learning glacier dynamics from data. With Bryan Riel (MIT)
2021	PI – MIT Climate Grand Challenges: Ice Sheet Stability and Rapid Sea-Level Rise: Exploring Risk, Impact, and Mitigation Strategies
2021	Researcher – MIT Climate Grand Challenges: Stratospheric Airborne Climate Observatory System to Initiate Revolution In Climate Risk Forecasting. With PI John Hansman (MIT)
2021–2022	Researcher – National Geographic Society: AI for Glacial Lake Outburst Floods Hazard Potential Assessment in Chitral, Pakistan (AI4GLOF). With PI Muhammad Adnan Siddique (Information Technology University, Lahore, Pakistan)
2020–2021	PI – JPL Strategic University Research Partnership: Ice sheet mechanical properties as revealed from time-varying surface velocity fields. Award number 1657974.
2020–2022	PI – NEC Corporation Fund for Research in Computers and Communications: Quantifying the Evolution of Stress Fields in Antarctic Ice Shelves by Fusing Data from Multiple Remote Sensing Platforms, Instruments, and Techniques.
2020–2023	PI – NSFGEO-NERC: Collaborative Research: A new mechanistic framework for modeling rift processes in Antarctic ice shelves validated through improved strain-rate and seismic observations. Award number 1853918
2020–2021	PI – Microsoft AI for Earth: Theory-guided discovery of glacier dynamics using deep learning. With Bryan Riel (MIT)
2020–2021	PI – MIT-Germany - FAU Seed Fund: Rift Formation in Antarctic Ice Shelves: Constraining the Mechanical Properties of Glacier Ice Using Time-Dependent, High-Resolution Strain-Rate Fields. With Matthias Braun (FAU)
2019–2020	PI – Earl A Killian III (1978) and Waidy Lee Fund: Improving projections of sea-level rise by (deep) learning glacier dynamics from data. With Bryan Riel (MIT)
2019–2020	Co-I – Microsoft AI for Earth: Theory-guided discovery of glacier dynamics using deep learning. With Bryan Riel (MIT) and Tobias Bischoff (Microsoft)
2018–2023	Researcher – NSF-NERC: Processes, drivers, predictions: Modeling the history and evolution of Thwaites Glacier (PROPHET). Award number 1739031
2016–2018	PI – Spatiotemporal characteristics of basal resistance to ice flow in the West Antarctic Ice Sheet from satellite observations and numerical modeling. NSF Earth Sciences Postdoctoral Fellowship award 1452587
2013–2015	Research assistant – Subglacial mechanics using repeat-pass InSAR measurements and numerical models of temperate ice caps in Iceland. NASA Cryospheric Science award NNX14AH80G, with PI Mark Simons (Caltech)
2011–2014	PI – Investigating the mechanics of subglacial till using airborne radar interferometry and numerical ice flow models. NASA Earth and Space Science Fellowship
2011–2012	Research assistant – Temperate glacier studies with UAVSAR. NASA Cryospheric Science, with PI Mark Simons (Caltech)

Mentorship

Postdoctoral researchers

2026–	Nicolas Morales Preciado (University of British Columbia PhD '26), starting 2026
2026–	Natasha Morgan-Witts (University of Wisconsin, Madison PhD '26), starting July 2026
2026–	George Lu (Columbia University PhD '26), starting October 2026
2025–	Evan Kramer (MIT PhD '25), Caltech Seismo Lab Director's Fellow
2018–2021	Elizabeth (Lizz) Ultee (University of Michigan PhD '18); now research scientist at NASA Goddard Space Flight Center and Morgan State University

Graduate students, as a primary advisor

2024–	Thatcher Chamberlin (PhD, MIT EAPS (PAOC))
2023–	Max Filter (PhD, MIT EAPS (Geophysics))
2023–2025	Sarah Wells-Moran (MIT MS '25), now PhD student at University of Chicago
2022–2024	Ufuoma Ovienmhada (MIT PhD '24)
2021–2023	Neosha Narayanan (MIT MS '23), now PhD student at Georgia Tech
2019–2025	Faye Hendley Elgart (MIT PhD '25), now postdoc at Laboratoire de Glaciologie at the Université libre de Bruxelles
2019–	Justin Linick (PhD, MIT EAPS (Geophysics))
2018–2023	Joanna Millstein (MIT-WHOI JP PhD '23), now postdoc at Colorado School of Mines
2018–2022	Fiona Clerc (MIT-WHOI JP PhD '22), now faculty at Stony Brook University
2018–2022	Meghana Ranganathan (MIT PhD '22), now faculty at University of Chicago
2018–2019	Brindha Kanniah (MIT MS '19)

Graduate students, as a secondary or project advisor

2025–	PhD project advisor: Jackson Fellows, Caltech GPS (Seismo Lab)
2025–	PhD project advisor: Mma Ikwut-Ukwa, Caltech GPS (Planetary)
2025–	PhD project advisor: Ben Reynolds, Caltech GPS (Geology)
2025–	PhD project advisor: Elena Dworak, Caltech GPS (ESE)
2025–	PhD project advisor: Yi (Lain) Li, Caltech GPS (Seismo Lab)
2024–	PhD project advisor: Caroline Needell, MIT-WHOI Joint Program
2023–	PhD project advisor: Alex Miller, MIT AeroAstro
2022–	PhD project advisor: Caroline Mouchon, MIT EAPS (Geophysics)
2022–2024	PhD project advisor: Annick Dewald (MIT PhD '24, AeroAstro), now founding faculty at Greenway Institute
2021–2023	PhD project advisor: Grace O'Neil, EAPS (PAOC)
2021–2023	PhD project advisor: Yudong Sun (MIT PhD '25), now postdoc at Stanford University
2021–2023	PhD project advisor: Mila Lubeck, EAPS (Geophysics)
2019–2020	MS project co-advisor: Pierce Hunter (UO MS '20)
2019–2024	PhD project advisor: Erik Tamre (MIT PhD '24), now postdoc at Dartmouth College
2018–2022	PhD project advisor: Kasturi Shah (MIT PhD '22), now McDonnell Foundation Research Fellow at University of Cambridge
2018–2019	PhD project advisor: Julia Wilcots (MIT PhD '22), now faculty at University of Minnesota, Twin Cities

Undergraduate students

(UROP = MIT Undergrad Research Opportunities Program; MSRP = MIT Summer Research Program)

2024	UROP advisor: Justin Cole, MIT
2023–2025	UROP advisor: Rina Cao, MIT
2023	UROP advisor: Kyle Guerre, MIT
2023	UROP advisor: Kristine Bridges, MIT
2021	MSRP advisor: Florencia Corbo-Ferreira, U. of Florida
2021	UROP advisor: Mateo Pisinger, MIT
2021–2022	UROP advisor: Ryan Conti, Math + Comp Sci, MIT
2021	UROP advisor: Jon Rosario, Math + Comp Sci, MIT
2020	UROP advisor: Neosha Narayanan, Mat. Sci., MIT
2020–2023	UROP advisor: Sarah Wells-Moran, Wellesley College
2020–2023	Project advisor: Max Filter, Computer Science, U. of Virginia
2020–2021	UROP advisor: Jack-William Barotta, Math, MIT
2020	UROP advisor: Gabriela Alvarez, Mech. Eng., MIT
2020	UROP advisor: Jordan Ambrosio, Mech. Eng., MIT
2020	UROP advisor: Stephanie Baez, Civil Eng., MIT
2020	UROP advisor: Anna Chau, EE + Comp. Sci, MIT
2020	UROP advisor: Adriana Flores, AeroAstro, MIT
2020	UROP advisor: Meriah Gannon, Env. Eng., MIT
2020	UROP advisor: Adelynn Paik, Sys. Eng., MIT
2020	UROP advisor: Kristopher Vu, Mech. Eng. + Comp. Sci, MIT
2020	UROP advisor: Joyce Yoon, Mech. Eng., MIT
2015	Project advisor: Benjamin Lauer, Université de Lorraine
2015	Project advisor: Christine Rains, DEVELOP-JPL
2015	Project advisor: Jerry Heo, DEVELOP-JPL
2015	Project advisor: Erika Higa, DEVELOP-JPL
2013	Project advisor: Sandia Akhtar, Caltech

Thesis Committee Membership

(in addition to mentorship)

Internal thesis committees

2023–	Riley Thomas (PhD, MIT-WHOI JP)
2023–	Maia Cohen (PhD, MIT-WHOI JP)
2023–	Caroline Needell (PhD, MIT-WHOI JP)
2023–2025	Evan Kramer (PhD, MIT AeroAstro)
2023–	Kristen Ammons (PhD, MIT AeroAstro)
2022	Mathilde Wimez (MS 2022, MIT EAPS)
2021–2023	Ana Glidden (PhD, MIT EAPS)
2021–2025	Alan Gaul (PhD, MIT-WHOI JP)
2021–2023	Cadence Payne (PhD, MIT AeroAstro)
2020	Taylor Safrit (MS 2020, MIT EAPS)
2020	Anuar Togaibekov (MS 2020, MIT EAPS)

2019	Dylan Cohen (MS 2019, MIT EAPS)
2018–2020	Jeffery Mei (PhD, MIT-WHOI JP; defended Aug 2020)
2018–2021	Eric Stansifer (PhD, MIT EAPS; defended Oct 2021)

External thesis committees

2022	Aaron Stubblefield (PhD 2022, Columbia University)
2021	Cornelius Quigley (PhD 2021, University of Tromso)
2019	Martine Espeseth (PhD 2019, University of Tromso)
2018–2021	Juan Pinales (MS 2021, University of Miami)
2018–2023	Stephanie Olinger (PhD 2023, Harvard)

Teaching

2026 Spring	Caltech Ge193	Glacier Dynamics
2025 Spring	MIT 12.421/12.621	Physical Principles of Remote Sensing (undergraduate and graduate)
2024 Spring	MIT 12.203/12.503	Mechanics of Earth (undergraduate and graduate)
2024 Spring	MIT 12.512	Field Geophysics Analysis (graduate)
2024 IAP	MIT 12.511	Geophysics Field Camp (graduate)
2023 Fall	MIT 12.202/12.502	Flow, Deformation, and Fracture in Earth and Other Terrestrial Bodies (undergraduate and graduate)
2023 Fall	MIT 12.214/12.507	Essentials of Applied Geophysics (undergraduate and graduate)
2022 Fall	MIT 12.421/12.621	Physical Principles of Remote Sensing (undergraduate and graduate)
2022 Spring	MIT 12.203/12.503	Mechanics of Earth (undergraduate and graduate)
2022 IAP	MIT 12.511	Geophysics Field Camp (graduate)
2021 Fall	MIT 12.S595	Geophysics Field Course (graduate)
2021 Fall	MIT 12.202/12.502	Flow, Deformation, and Fracture in Earth and Other Terrestrial Bodies (undergraduate and graduate)
2021 Spring	MIT 12.S595	Antarctic Ice Sheet in a Changing Climate (graduate)
2020 Fall	MIT 12.421/12.621	Physical Principles of Remote Sensing (undergraduate and graduate)
2020 Spring	MIT 12.203/12.503	Mechanics of Earth (undergraduate and graduate)
2019 Fall	MIT 12.202/12.502	Flow, Deformation, and Fracture in Earth and Other Terrestrial Bodies (undergraduate and graduate)
2019 Spring	MIT 12.005/12.520	Applications of Continuum Mechanics to Earth, Atmospheric, and Planetary Sciences (undergraduate) / Geodynamics (graduate)
2018 Fall	MIT 12.421/12.621	Physical Principles of Remote Sensing (undergraduate and graduate)

Academic Service

(internal and external)

2024–2025	MIT EAPS	Chair, Program in Geophysics
2024	MIT EAPS	Department Head search committee
2024	MIT EAPS	Graduate admissions committee
2023	MIT EAPS	Renovation committee (chair)
2023–2025	MIT-WHOI	MIT-WHOI Joint Program Committee
2022–2025	MIT	Warrior Scholar Project
2022–2023	MIT EAPS	Department website committee
2021	UNAVCO	Guest lecturer for InSAR/ISCE workshop
2021–2024	UNAVCO	WInSAR Executive Committee member
2021–2022	MIT EAPS	Department Lecture Series committee
2020–2024	MIT	Common Ground for Computing Education Committee
2020	NASA	Surface Deformation and Change – Cryosphere Working Group
2019–2025	AGU Journals	Associate Editor, Journal of Geophysical Research: Earth Surface
2019–2020	MIT EAPS	Task Force 2023 committee member
2018–2025	MIT EAPS	Communications committee member
2018–2021	Boston Museum of Science	External advisory committee member
2018–2019	MIT EAPS	Faculty search committee member

Professional Activities

Chief Scientist and Co-Founder: Arête Glacier Initiative (2024–)
Associate Editor: Journal of Geophysical Research: Earth Surface (2019–2025)
Scientific Editor: Annals of Glaciology (2017–2018)
Member: International Glaciological Society, International Association of Cryospheric Sciences, Association of Polar Early Career Scientists, American Geophysical Union, European Geosciences Union, American Association for the Advancement of Science
Notable external committees: Boston Museum of Science advisory board member; NASA Surface Deformation and Change – Cryosphere Working Group; UNAVCO WInSAR Executive Committee member

Selected Outreach Efforts

– Faculty Advisor, MIT Student Veterans Association
– Speaker and Lecturer, Warrior Scholar Project
– Speaker, Lowell Observatory public science event: I Heart Pluto Festival
– Military veteran advising and speaking
– MIT Museum: Teen Science Café
– Boston Museum of Science *Pulsar* science podcast
– Gardner Pilot Academy, Allston, MA: Introducing 8th grade math students to glaciers and glaciology
– *Arctic Adventure: Exploring with Technology* at the Boston Museum of Science
– NASA Climate Day presenter
– US Embassy Iceland outreach on Arctic climate

– Iridescent Learning: Teaching children about scientific concepts

Military Service

United States Marine Corps
Active duty August 1996–February 2004
Rank: Staff Sergeant
Units: HMX-1, HMH-461, HMM-264, 26th MEU (aboard USS Iwo Jima LHD-7)
Major campaigns: Operation Iraqi Freedom, Combined Joint Task Force–Horn of Africa, Joint Task Force Liberia